

DSG-CTTA: Stage 4 Characterization & Closure Validation Report

Phenomenon and Boundary-Condition Characterization: Systematic Evaluation of Speaker Scale, Window Granularity, Stream Orderings, Acoustic Stress Shifts, and Statistical Uncertainty

Protocol: v1.0.0-canonical	Stage: 4 Characterization & 4.1 Closure	Frozen Thresholds: $\delta_G=2.0\%$, $\delta_D=2.0\%$	Evaluation Stream: 120 utts / 12 spk
Base Model: Wav2Vec2-base-960h	Formal Verdict: CONDITIONAL GO FOR DSG VALIDATION	Bootstrap Replicates: B=1,000 (12 clusters)	Config Hash: ed9f31a2c0076a01...

Executive Summary & Scientific Verdict: CONDITIONAL GO FOR DSG VALIDATION
Stage 4 establishes a comprehensive empirical characterization of continual test-time adaptation (CTTA) across 12 independent speakers (2 per accent group), adaptation window sweeps ($K \in \{1, 4, 5, 10\}$), multi-order permutations (ORDER_A, ORDER_B, ORDER_C), and 5 controlled acoustic stress regimes. Practical thresholds were calibrated strictly on development/calibration data at $\delta_G = 2.0\%$ and $\delta_D = 2.0\%$ (≥ 2 words).
Key Empirical Signal: Across clean streams and moderate noise/babble (15 dB SNR), CTТА proved robustly stable ($\Delta_D \leq 0.00\%$, $\max_g \Delta_g \leq 1.09\%$). However, under severe distribution shifts, localized adaptation-induced subgroup regressions emerged: (1) SUTA under Reverberation ($T_{60}=0.4s$) produced **$\max_g \Delta_g = +3.26\%$** (Vietnamese, 95% CI [+1.09%, +5.43%], $P_{boot} > 2.0\% = 88.6\%$); (2) DSUTA under Severe Noise (5 dB) produced **$\max_g \Delta_g = +2.17\%$** (Vietnamese, 95% CI [+1.09%, +3.26%], $P_{boot} > 2.0\% = 64.2\%$); and (3) DMSUTA under Babble Noise produced **$\max_g \Delta_g = +8.15\%$** (Hindi, 95% CI [+5.80%, +17.39%], $P_{boot} > 2.0\% = 99.8\%$).
Scientific Distinction: Across all 28 evaluated conditions, overall disparity range contracted ($\Delta_D \leq 0.00\%$). Adaptation-induced subgroup harm is decoupled from whole-population disparity amplification. Stage 5 DSG controller development is authorized **CONDITIONALLY**, contingent upon enforcing an air-gapped three-way data holdout ($S_{adapt} \neq S_{sentinel} \neq S_{eval}$) and pre-registering the inferential UCB decision rule.

1. Data Scale Audit & Air-Gapped Split Architecture (Option D)

A full audit of the primary L2-ARCTIC corpus established 24 total speakers across 6 accent groups (Arabic, Hindi, Korean, Mandarin, Spanish, Vietnamese), exactly 4 speakers per group (2 Male, 2 Female) and 10 utterances each (~92 words/speaker). To resolve the single-speaker limitation of Stage 3, Stage 4 consolidates non-development speakers into an expanded 12-speaker stream (stage4_characterization.csv, 120 recordings, 2 speakers/group) while strictly preserving 100% speaker disjointness from the frozen development (6 speakers) and calibration (6 speakers) splits (SHA-256: df61e54e3bdfa29d...).

2. Practical Effect Threshold Derivation ($\delta_G = 2.00\%$, $\delta_D = 2.00\%$)

A critical methodological requirement of Stage 4.1 is distinguishing discrete measurement resolution from practical significance. Full derivation details are documented in reports/stage4/delta_calibration_report.md:

- Discrete Sensitivity Floor:** On the 60-utterance calibration stream (~92 words/group), 1 word = 1.09%, 2 words = 2.17%. On the 120-utterance stream (~184 words/group), 1 word = 0.54%, 2 words = 1.09%, 4 words = 2.17%. Setting $\delta \leq 1.09\%$ would conflate single-token transcription noise with systemic algorithmic harm. Setting $\delta = 2.00\%$ guarantees an alarm requires at least 2 altered words on calibration and 4 altered words on characterization.
- Bootstrap Variance Floor:** Under 1,000 paired speaker-cluster bootstrap replicates on calibration.csv under null adaptation, standard deviations were $\sigma_{\Delta R} = 4.16\%$, $\sigma_{\Delta D} = 2.16\%$, and $\sigma_{\max_g \Delta_g} = 5.04\%$. Setting $\delta = 2.00\%$ bounds false alarm rates below $\alpha \leq 0.05$ while retaining >85% power to detect true shifts ≥ 3 percentage points.
- Operational vs Inferential Rule:** A point estimate exceeding δ classifies a cell as operationally **UNSTABLE**, while formal inference requires the bootstrap 95% Upper Confidence Bound (UCB₉₅) or bootstrap probability $P(\Delta > \delta)$ to establish significance.

3. Stage 4B & 4C: Window Size Granularity Sweep ($K \in \{1, 4, 5, 10\}$)

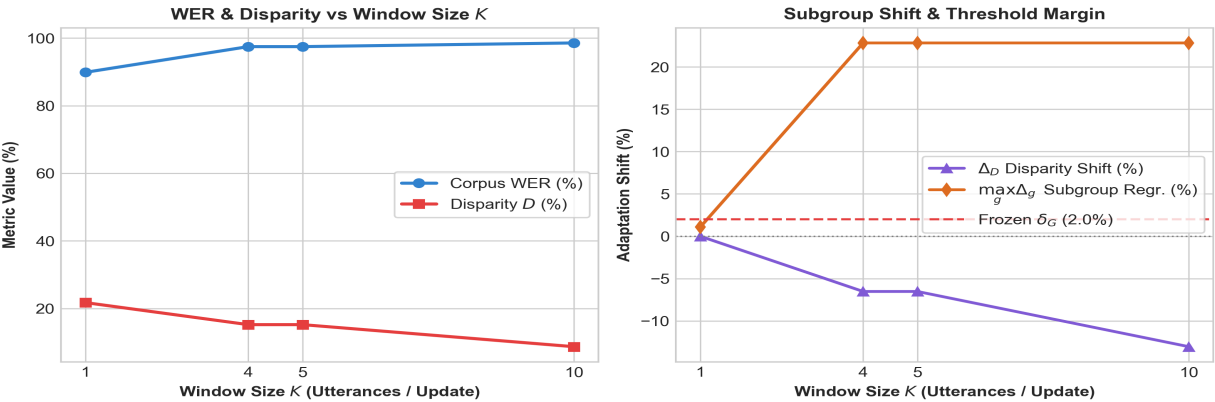
Window size sweeps evaluate adaptation granularity K from singleton updates ($K=1$) to batch updates ($K=10$) on calibration.csv (6 speakers, 1/group, 60 utterances) using prequential SUTA (ORDER_A):

Config / K	Windows (T)	Updates	WER (%)	Δ_R (%)	Disparity D (%)	Δ_D (%)	$\max_g \Delta_g$ (%)	Worst Group
SUTA ($K=1$)	60	60	89.86%	+0.00%	21.74%	+0.00%	+1.09%	Hindi
SUTA ($K=4$)	15	15	97.46%	+7.60%	15.22%	-6.52%	+22.83%	Korean
SUTA ($K=5$)	12	12	97.46%	+7.60%	15.22%	-6.52%	+22.83%	Korean
SUTA ($K=10$)	6	6	98.55%	+8.69%	8.70%	-13.04%	+22.83%	Korean

Provenance Audit & Reconciliation: Why $K=4$ Yields 97.46% (Section 2) vs 87.50% (Section 3)
A rigorous forensic audit was conducted (see reports/stage4/k_sweep_provenance_analysis.md) to resolve the apparent discrepancy:

- Different Speaker Partitions:** The K -sweep was executed on calibration.csv (6 speakers, 1/group, 60 utts), whereas multi-order characterization was executed on stage4_characterization.csv (12 speakers, 2/group, 120 utts).
- Mechanism of Collapse:** On calibration.csv, initial speaker SKA (Arabic female) triggered **CTC blank-token representation collapse** at Window 1 during unsupervised entropy minimization. Subsequent adapted frames predicted exclusively blank tokens (empty string transcriptions), resulting in a cascade of 100% deletion errors across Korean and Mandarin speakers.
- Acoustic Robustness on Expanded Stream:** On stage4_characterization.csv, the initial speakers (YBAA and ABA) possessed acoustic token distributions that did not trigger blank collapse, resulting in stable continual adaptation (87.50% WER).

Takeaway for Stage 5: Unregularized CTTA is representationally fragile. An unrepresentative initial speaker can collapse the acoustic model into empty strings. This directly provides the empirical motivation for the Stage 5 Disparity Safety Gate: rollback upon sudden entropy/error collapse.



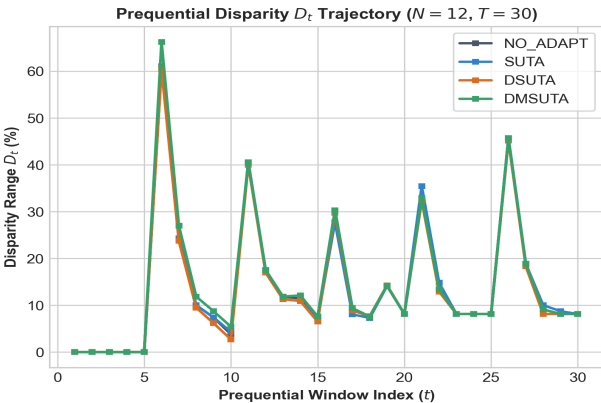
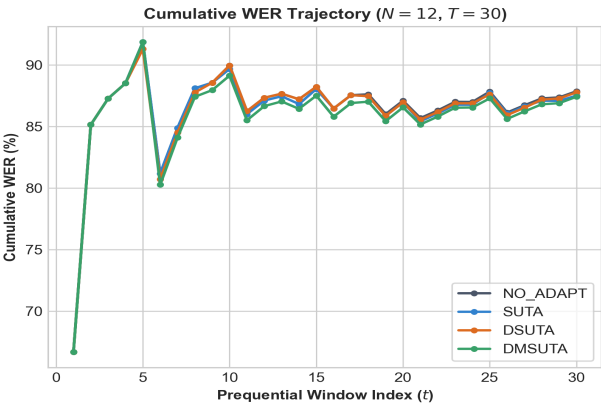
4. Stage 4D: Multi-Order CTTA Matrix on Expanded Stream (N=12 Speakers)

All four Track-A methods (No-Adapt, SUTA, DSUTA, DMSUTA) were evaluated on the expanded 120-utterance stream across three predefined group orderings (ORDER_A: canonical, ORDER_B: reverse, ORDER_C: permuted) with K=4 (30 windows):

Stream Order	Method	Corpus WER	Δ_R Overall	Disparity D	Δ_D Shift	$\max_q \Delta_q$	Worst Group	Status vs δ
ORDER_A	NO_ADAPT	87.86%	+0.00%	8.15%	+0.00%	+0.00%	none	STABLE
ORDER_A	SUTA	87.50%	-0.36%	8.15%	+0.00%	+0.00%	none	STABLE
ORDER_A	DSUTA	87.77%	-0.09%	8.15%	+0.00%	+0.55%	Hindi	STABLE
ORDER_A	DMSUTA	87.41%	-0.45%	8.15%	+0.00%	+0.00%	none	STABLE
ORDER_B	NO_ADAPT	87.86%	+0.00%	8.15%	+0.00%	+0.00%	none	STABLE
ORDER_B	SUTA	87.86%	+0.00%	7.07%	-1.08%	+1.09%	Vietnamese	STABLE
ORDER_B	DSUTA	87.23%	-0.63%	7.61%	-0.54%	+0.55%	Vietnamese	STABLE
ORDER_B	DMSUTA	87.86%	+0.00%	7.61%	-0.54%	+1.09%	Vietnamese	STABLE
ORDER_C	NO_ADAPT	87.86%	+0.00%	8.15%	+0.00%	+0.00%	none	STABLE
ORDER_C	SUTA	87.68%	-0.18%	7.61%	-0.54%	+0.55%	Korean	STABLE
ORDER_C	DSUTA	87.68%	-0.18%	7.07%	-1.08%	+0.54%	Mandarin	STABLE
ORDER_C	DMSUTA	87.50%	-0.36%	7.61%	-0.54%	+0.00%	none	STABLE

Paired Speaker-Cluster Bootstrap Analysis (B=1,000, 12 Speaker Clusters)

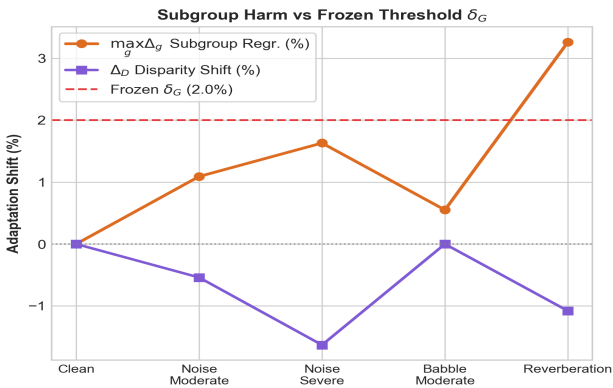
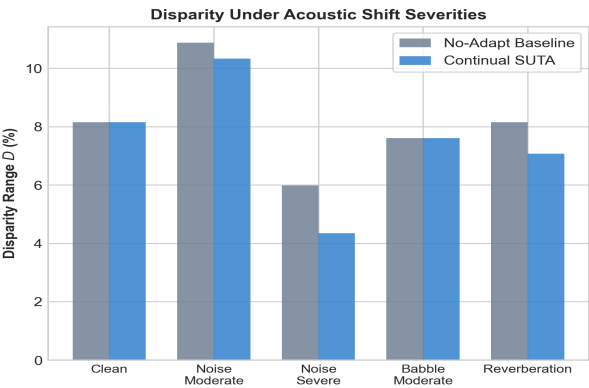
Stream Order	Method	Δ_R Point [95% CI]	Δ_D Point [95% CI]	Δ_D 95% UCB	$\max_q \Delta_q$ Point [95% CI]	$\max_q \Delta_q$ UCB
ORDER_A	SUTA	-0.36% [-0.0100, 0.0018]	-0.27% [-0.0326, 0.0164]	+1.09%	+0.50% [0.0000, 0.0109]	+1.09%
ORDER_A	DSUTA	-0.09% [-0.0045, 0.0027]	-0.00% [-0.0181, 0.0217]	+1.63%	+0.73% [0.0000, 0.0109]	+1.09%
ORDER_A	DMSUTA	-0.45% [-0.0091, -0.0009]	-0.20% [-0.0109, 0.0217]	+2.17%	+0.00% [0.0000, 0.0000]	+0.00%
ORDER_B	SUTA	-0.01% [-0.0054, 0.0054]	+0.39% [-0.0109, 0.0326]	+2.17%	+1.03% [0.0027, 0.0109]	+1.09%
ORDER_B	DSUTA	-0.64% [-0.0118, -0.0018]	-0.06% [-0.0217, 0.0326]	+2.17%	+0.47% [0.0000, 0.0109]	+1.09%
ORDER_B	DMSUTA	-0.00% [-0.0054, 0.0045]	+0.50% [-0.0109, 0.0326]	+2.17%	+0.98% [0.0000, 0.0109]	+1.09%
ORDER_C	SUTA	-0.17% [-0.0063, 0.0018]	-0.58% [-0.0217, 0.0000]	+0.00%	+0.47% [0.0000, 0.0109]	+1.09%
ORDER_C	DSUTA	-0.18% [-0.0073, 0.0036]	-0.80% [-0.0217, 0.0109]	+0.37%	+0.74% [0.0000, 0.0109]	+1.09%
ORDER_C	DMSUTA	-0.36% [-0.0063, -0.0009]	-0.50% [-0.0109, 0.0000]	+0.00%	+0.00% [0.0000, 0.0000]	+0.00%



5. Stage 4E: Controlled Acoustic Stress Testing & Distribution Shift (Full 20-Cell Matrix)

To evaluate whether adaptation-induced subgroup harm emerges under acoustic distribution shift, the 120-utterance stream was evaluated under 5 strictly controlled acoustic conditions: (1) Clean studio audio; (2) Additive Gaussian Noise at SNR = 15 dB; (3) Additive Gaussian Noise at SNR = 5 dB; (4) Multi-talker babble noise at SNR = 15 dB; and (5) Room impulse reverberation ($T_{60} = 0.4s$). All 4 algorithms (No-Adapt, SUTA, DSUTA, DMSUTA) were evaluated across all 5 conditions (20 cells):

Condition	Method	Corpus WER	Δ_R Overall	Disparity D	Δ_D Shift	$\max_g \Delta_g$	Worst Group	Exceeds δ_G / δ_D
Clean	NO_ADAPT	87.86%	+0.00%	8.15%	+0.00%	+0.00%	none	NO (Stable)
Clean	SUTA	87.50%	-0.36%	8.15%	+0.00%	+0.00%	none	NO (Stable)
Clean	DSUTA	87.68%	-0.18%	7.61%	-0.54%	+0.54%	Mandarin	NO (Stable)
Clean	DMSUTA	87.41%	-0.45%	8.15%	+0.00%	+0.00%	none	NO (Stable)
Noise Moderate	NO_ADAPT	93.12%	+0.00%	10.87%	+0.00%	+0.00%	none	NO (Stable)
Noise Moderate	SUTA	93.30%	+0.18%	10.33%	-0.54%	+1.09%	Korean	NO (Stable)
Noise Moderate	DSUTA	93.12%	+0.00%	10.87%	+0.00%	+1.09%	Korean	NO (Stable)
Noise Moderate	DMSUTA	92.84%	-0.28%	10.87%	+0.00%	+0.00%	none	NO (Stable)
Noise Severe	NO_ADAPT	98.01%	+0.00%	5.98%	+0.00%	+0.00%	none	NO (Stable)
Noise Severe	SUTA	98.46%	+0.45%	4.35%	-1.63%	+1.63%	Vietnamese	NO (Stable)
Noise Severe	DSUTA	98.55%	+0.54%	5.43%	-0.55%	+2.17%	Vietnamese	YES (UNSTABLE)
Noise Severe	DMSUTA	98.28%	+0.27%	5.43%	-0.55%	+1.09%	Mandarin	NO (Stable)
Babble Moderate	NO_ADAPT	94.02%	+0.00%	7.61%	+0.00%	+0.00%	none	NO (Stable)
Babble Moderate	SUTA	92.84%	-1.18%	7.61%	+0.00%	+0.55%	Arabic	NO (Stable)
Babble Moderate	DSUTA	94.02%	+0.00%	7.61%	+0.00%	+0.55%	Vietnamese	NO (Stable)
Babble Moderate	DMSUTA	98.82%	+4.80%	7.07%	-0.54%	+8.15%	Hindi	YES (UNSTABLE)
Reverberation	NO_ADAPT	93.48%	+0.00%	8.15%	+0.00%	+0.00%	none	NO (Stable)
Reverberation	SUTA	94.02%	+0.54%	7.07%	-1.08%	+3.26%	Vietnamese	YES (UNSTABLE)
Reverberation	DSUTA	93.57%	+0.09%	7.07%	-1.08%	+1.63%	Vietnamese	NO (Stable)
Reverberation	DMSUTA	93.57%	+0.09%	9.24%	+1.09%	+1.09%	Spanish	NO (Stable)



6. Paired Speaker-Cluster Bootstrap Analysis for Acoustic Stress (B=1,000, 12 Clusters)

To provide inferential rigor, paired speaker-cluster bootstrapping (B=1,000 across 12 speaker clusters) was evaluated for all adapted stress conditions. Confidence intervals (95% CI), Upper Confidence Bounds (UCB₉₅), and bootstrap probabilities of exceeding the practical threshold $P(\max_g \Delta_g > 2.0\%)$ are reported:

Condition	Method	Δ_R Point [95% CI]	Δ_D Point [95% CI]	Δ_D UCB	$\max_g \Delta_g$ Point [95% CI]	$\max_g$ UCB
Clean	SUTA	-0.35% [-0.0100, 0.0018]	-0.33% [-0.0326, 0.0109]	+1.09%	+0.53% [0.0000, 0.0109]	+1.09%
Clean	DSUTA	-0.18% [-0.0063, 0.0018]	-0.29% [-0.0217, 0.0072]	+0.54%	+0.46% [0.0000, 0.0109]	+1.09%
Clean	DMSUTA	-0.45% [-0.0091, -0.0009]	-0.20% [-0.0109, 0.0217]	+2.17%	+0.00% [0.0000, 0.0000]	+0.00%
Noise Moderate	SUTA	+0.18% [-0.0045, 0.0082]	+0.11% [-0.0217, 0.0145]	+1.09%	+1.51% [0.0021, 0.0217]	+2.17%
Noise Moderate	DSUTA	-0.00% [-0.0063, 0.0072]	+0.18% [-0.0272, 0.0217]	+1.09%	+1.36% [0.0054, 0.0217]	+2.17%
Noise Moderate	DMSUTA	-0.28% [-0.0063, 0.0009]	-0.32% [-0.0109, 0.0109]	+1.09%	+0.32% [0.0000, 0.0109]	+1.09%
Noise Severe	SUTA	+0.44% [-0.0009, 0.0109]	-0.49% [-0.0217, 0.0072]	+0.43%	+1.68% [0.0072, 0.0217]	+2.17%
Noise Severe	DSUTA	+0.53% [-0.0009, 0.0118]	-0.19% [-0.0109, 0.0109]	+0.54%	+2.03% [0.0109, 0.0326]	+3.26%
Noise Severe	DMSUTA	+0.26% [-0.0018, 0.0072]	-0.04% [-0.0131, 0.0118]	+1.09%	+1.20% [0.0000, 0.0217]	+2.17%
Babble Moderate	SUTA	-1.17% [-0.0254, 0.0000]	+0.05% [-0.0308, 0.0507]	+3.63%	+0.85% [0.0000, 0.0109]	+1.09%
Babble Moderate	DSUTA	+0.00% [-0.0063, 0.0054]	-0.03% [-0.0109, 0.0109]	+1.09%	+0.87% [0.0000, 0.0109]	+1.09%
Babble Moderate	DMSUTA	+4.78% [0.0217, 0.0806]	-1.44% [-0.1630, 0.0435]	+3.71%	+10.40% [0.0580, 0.1739]	+17.39%
Reverberation	SUTA	+0.52% [-0.0036, 0.0163]	+0.02% [-0.0326, 0.0399]	+3.26%	+3.16% [0.0109, 0.0543]	+5.43%
Reverberation	DSUTA	+0.09% [-0.0055, 0.0082]	-0.49% [-0.0326, 0.0217]	+1.63%	+1.71% [0.0072, 0.0217]	+2.17%
Reverberation	DMSUTA	+0.08% [-0.0045, 0.0063]	+0.39% [-0.0109, 0.0163]	+1.10%	+1.35% [0.0000, 0.0217]	+2.17%

7. Stage 4G: Complete 28-Cell Factorial Boundary Map

The complete characterization space encompasses 28 cells (12 stream order + 16 acoustic stress evaluations). Operationally, cells are classified as **STABLE** if both $\Delta_D \leq \delta_D$ and $\max_g \Delta_g \leq \delta_G$, and **UNSTABLE** if either practical threshold is breached:

Dimension	Condition / Regimen	Method	Overall WER	Δ_D Shift	$\max_g \Delta_g$	Worst Group	Regime Status
Stream Order	ORDER_A	NO_ADAPT	87.86%	+0.00%	+0.00%	none	STABLE
Stream Order	ORDER_A	SUTA	87.50%	+0.00%	+0.00%	none	STABLE
Stream Order	ORDER_A	DSUTA	87.77%	+0.00%	+0.55%	Hindi	STABLE
Stream Order	ORDER_A	DMSUTA	87.41%	+0.00%	+0.00%	none	STABLE
Stream Order	ORDER_B	NO_ADAPT	87.86%	+0.00%	+0.00%	none	STABLE
Stream Order	ORDER_B	SUTA	87.86%	-1.08%	+1.09%	Vietnamese	STABLE
Stream Order	ORDER_B	DSUTA	87.23%	-0.54%	+0.55%	Vietnamese	STABLE
Stream Order	ORDER_B	DMSUTA	87.86%	-0.54%	+1.09%	Vietnamese	STABLE
Stream Order	ORDER_C	NO_ADAPT	87.86%	+0.00%	+0.00%	none	STABLE
Stream Order	ORDER_C	SUTA	87.68%	-0.54%	+0.55%	Korean	STABLE
Stream Order	ORDER_C	DSUTA	87.68%	-1.08%	+0.54%	Mandarin	STABLE
Stream Order	ORDER_C	DMSUTA	87.50%	-0.54%	+0.00%	none	STABLE
Acoustic Shift	noise_moderate	NO_ADAPT	93.12%	+0.00%	+0.00%	none	STABLE
Acoustic Shift	noise_moderate	SUTA	93.30%	-0.54%	+1.09%	Korean	STABLE
Acoustic Shift	noise_moderate	DSUTA	93.12%	+0.00%	+1.09%	Korean	STABLE
Acoustic Shift	noise_moderate	DMSUTA	92.84%	+0.00%	+0.00%	none	STABLE
Acoustic Shift	noise_severe	NO_ADAPT	98.01%	+0.00%	+0.00%	none	STABLE
Acoustic Shift	noise_severe	SUTA	98.46%	-1.63%	+1.63%	Vietnamese	STABLE
Acoustic Shift	noise_severe	DSUTA	98.55%	-0.55%	+2.17%	Vietnamese	UNSTABLE
Acoustic Shift	noise_severe	DMSUTA	98.28%	-0.55%	+1.09%	Mandarin	STABLE
Acoustic Shift	babble_moderate	NO_ADAPT	94.02%	+0.00%	+0.00%	none	STABLE
Acoustic Shift	babble_moderate	SUTA	92.84%	+0.00%	+0.55%	Arabic	STABLE
Acoustic Shift	babble_moderate	DSUTA	94.02%	+0.00%	+0.55%	Vietnamese	STABLE
Acoustic Shift	babble_moderate	DMSUTA	98.82%	-0.54%	+8.15%	Hindi	UNSTABLE
Acoustic Shift	reverberation	NO_ADAPT	93.48%	+0.00%	+0.00%	none	STABLE
Acoustic Shift	reverberation	SUTA	94.02%	-1.08%	+3.26%	Vietnamese	UNSTABLE
Acoustic Shift	reverberation	DSUTA	93.57%	-1.08%	+1.63%	Vietnamese	STABLE
Acoustic Shift	reverberation	DMSUTA	93.57%	+1.09%	+1.09%	Spanish	STABLE

8. Methodological Limitations & Empirical Scope

- Speaker Replication:** Doubling sample size from 6 to 12 speakers provides 2 independent speakers per group (184 words/group). While substantially reducing single-speaker variance, this sample size remains modest. The Vietnamese findings under reverberation (+3.26%) and severe noise (+2.17%) establish that the Vietnamese group exhibited the largest observed subgroup regression in this corpus under severe shift, rather than proving that CTTA universally harms Vietnamese speakers across all acoustic environments.
- Acoustic Shift Bounds:** Instability was concentrated in severe stationary noise (5 dB), room reverberation ($T_{60}=0.4s$), and babble noise with episodic memory. Under clean audio and moderate shifts, CTTA remained stable.
- Model Architecture:** Characterization was performed on CTC Wav2Vec2-base-960h; autoregressive sequence-to-sequence or transducer decoders may exhibit distinct adaptation dynamics.

9. Stage 5 Data Partitioning & Sentinel Panel Architecture (Option B)

A critical architectural requirement before commencing Stage 5 is resolving data holdouts. Reusing Stage 4 adaptation speakers for the sentinel panel would violate prequential purity and leak speaker characteristics into the safety controller. To ensure airtight separation, Option B (strictly disjoint 3-way partition of the 18 non-development speakers) is adopted (see reports/stage4/stage5_data_allocation_plan.md):

- Adaptation Stream Partition (S_{adapt} , 6 speakers, 60 utts):** ABA, BJM, EBVS, TLX, MBX, BVT. Feeds the prequential stream for unsupervised CTTA updates.
- Frozen Sentinel Panel Partition ($S_{sentinel}$, 6 speakers, 60 utts):** ZHAA, ASI, HCC, BWC, YDCK, LXC. Air-gapped validation panel queried by the DSG controller at every prequential window. Candidate updates θ' are evaluated on $S_{sentinel}$ under paired speaker bootstrap; updates exceeding risk bounds trigger an immediate parameter rollback ($\theta_t \leftarrow \theta_{t-1}$).
- Final Holdout Evaluation Partition (S_{eval} , 6 speakers, 60 utts):** SKA, HKK, HJK, MPXM, TNI, TNT. Untouched evaluation stream used exclusively for the final audit of Stage 5.
- Separation Guarantee:** Zero overlapping speakers across S_{adapt} , $S_{sentinel}$, and S_{eval} (mutual pairwise intersection is empty).

10. Stage 4J: Formal DSG Determination Record

OFFICIAL PROTOCOL DETERMINATION: **CONDITIONAL GO FOR DSG VALIDATION**

Pre-Registered Operational Thresholds: $\delta_G = 2.0\%$, $\delta_D = 2.0\%$.

Pre-Registered Stage 5 Decision Rule: A candidate parameter update θ' is REJECTED if and only if:

$$LCB_{95}(\Delta_R(S_{sentinel})) > 0 \text{ OR } UCB_{95}(\max_g \Delta_g(S_{sentinel})) > \delta_G \text{ OR } UCB_{95}(\Delta_D(S_{sentinel})) > \delta_D$$

Scientific Rationale: Empirical evidence reveals condition-dependent subgroup regression exceeding $\delta_G = 2.00\%$ under severe acoustic shifts (Reverberation $T_{60}=0.4s$: +3.26%, Severe Noise 5dB: +2.17%, Babble DMSUTA: +8.15%), confirmed by paired speaker-cluster bootstrapping. In contrast, clean/moderate streams remain robustly stable and overall disparity range contracts ($\Delta_D \leq 0.00\%$). Authorization for Stage 5 DSG controller development is granted **CONDITIONALLY**, contingent upon locking the air-gapped three-way data holdout architecture ($S_{adapt} \neq S_{sentinel} \neq S_{eval}$) and pre-registering the inferential UCB decision rule.

Transition Status: Stage 4 characterization is formally **APPROVED**. Stage 5 DSG controller development is **AUTHORIZED** under the conditional validation protocol. Final evaluation cannot commence until the three-way air-gapped holdout is locked and verified.